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A Deterministic Game-Theoretic Handover Protocol for UAV-RSU Integrated Networks

ZEESHAN SHAFIQ¹, MUHAMMAD ISMAIL², MOHAMMED ALI ALSHARA^{3,4},
TARIQULLAH JAN³, NAVEED AHMAD³, RUHUL AMIN KHALIL⁵ (*Member, IEEE*)

¹Center for Intelligent Systems and Networks Research, University of Engineering and Technology, Peshawar 25120, Pakistan.

²Faculty of Science, Engineering and Built Environment/School of Information Technology, Deakin University, Waurn Ponds VIC 3216, Australia.

³College of Computer and Information Sciences, Prince Sultan University, Riyadh 12435, Saudi Arabia.

⁴College of Computer and Information Sciences, Imam Mohammad Ibn Saud Islamic University, Riyadh 13318, Saudi Arabia.

⁵Engineering Requirement Unit (ERU), College of Engineering, United Arab Emirates University, Al-Ain 15551, UAE.

CORRESPONDING AUTHOR: Ruhul Amin Khalil (e-mail: ruhulamin@uaeu.ac.ae)

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ABSTRACT The increasing demand for time-sensitive applications in Intelligent Transportation Systems (ITS) powered by Vehicular Ad Hoc Networks (VANETs) burdens the limited communication channels available to the Road Side Units (RSUs), particularly in their dense serving areas. One of the most widely used applications of Unmanned Aerial Vehicles (UAVs) is to leverage them as on-demand aerial base stations to offload RSUs and provide a superior air-to-ground Line-of-Sight (LoS) channel. However, managing resource allocation between RSUs and UAVs in a highly dynamic vehicular network requires a novel predictive strategy. Current systems fail to jointly optimize UAV trajectory, channel resource allocation between the RSU and the UAV, and handover decisions between them to meet ultra-low-latency constraints. This paper proposes the Deterministic Coordinated Trajectory and Resource (DCRT) Optimization Framework, comprising a Kalman Filter for predictive vehicle-state estimation and a Cooperative Game Theory approach for optimal handover decisions between the RSU and the UAV. The DCRT framework solves a joint optimization problem to minimize total service time while maximizing average area throughput. Simulation results demonstrate the effectiveness of DCRT, showing an average service time reduction of approximately 61% compared with uncoordinated random access (RAC) and 48% compared with probabilistic access schemes (PAS) under medium-to-high traffic loads, while achieving a maximum throughput improvement of approximately **11.7% over RAC** and **34.6% over PAS** in terms of successful vehicles per slot under medium traffic densities. These results validate the deterministic game-theoretic approach for reliable ultra-low-latency V2X communication.

INDEX TERMS UAV Networks, Vehicular Ad Hoc Networks (VANETs), Resource Allocation, Trajectory Optimization, Game Theory, Handover, Ultra-Low Latency.

I. Introduction

THE integration of Unmanned Aerial Vehicles (UAVs) with conventional Road-Side Units (RSUs) in a dense environment can meet escalating demands of Vehicular Ad Hoc Networks (VANETs) [1]–[3]. As the vehicle density increases, especially during peak traffic hours, RSUs experience significant traffic and channel congestion, leading to reduced throughput and significant delays for time-sensitive applications [4], [5]. The development of novel resource allo-

cation and mobility management strategies can benefit this hybrid infrastructure, particularly for latency-sensitive and ultra-reliable services [2], [6]. Infrastructure-based VANETs, typically relying on RSUs operating under standards such as IEEE 802.11p, possess limited capacity due to a limited number of control and data communication channels, often resulting in their exhaustion, causing network congestion in high-density scenarios [1], [7]. In such cases, UAVs can provide on-demand services as mobile aerial base stations [1].

This augmentation is highly advantageous because the air-to-ground (G-to-A) channel established by UAVs inherently benefits from improved Line-of-Sight (LoS) probability and more favorable path loss components compared to the traditional ground-to-ground (G-to-G) channel [1], [8], [9]. However, integrating a mobile infrastructure component (the UAV) into a highly dynamic vehicular environment introduces a non-trivial problem of achieving ultra-low latency with constraints of higher throughput while tackling complex resource and trajectory coordination challenges and jointly coordinating UAV trajectory, radio resources, and handover processes [10]–[12].

There has been a number of research works conducted for UAV-assisted V2X communication, primarily focusing on resource-allocation (power, bandwidth, channel, computation) and/or trajectory or deployment problems independently, without delving into deterministic handover, game-theoretic design, or predictive vehicle-state-aware control [1], [13]–[16]. Few of them focused on joint resource allocation with UAV deployment or placement (trajectory-like but often static), relying on static or fixed-trajectory UAV deployment primarily for capacity offloading [2], [17]–[19]. More recently, studies have introduced mobility management, commonly employing heuristic algorithms or simple threshold-based mechanisms to trigger handover [1], [3], [20]–[22]. While existing air-ground and UAV-assisted VANET/V2X schemes do improve coverage and capacity by augmenting static RSU infrastructure with aerial or additional edge resources [1], [2], [13], [14], [17], [23]–[29], they typically rely on decomposed or RL-based reactive controllers for trajectory planning, association, and resource allocation, without explicit predictive models of vehicle mobility or channel evolution and without providing deterministic ultra-low-latency guarantees in the URLLC sense [17], [25]–[28]. Association and mobility management are generally handled via matching or selection [25], DRL/MARL policies [17], [26], [27], or simple LoS/SINR-threshold-based rules [1], [3], [20], rather than through predictive, game-theoretic handover mechanisms with explicit hysteresis, time-to-trigger, and interruption modeling. Treating trajectory control, mobility management, and radio/MEC resource allocation in such isolated or loosely coupled ways overlooks their fundamental interdependence, creating a critical gap in achieving truly seamless, anticipatory handover in high-mobility UAV-assisted vehicular environments.

Infrastructure-based VANETs relying solely on static RSUs often suffer from channel exhaustion, congestion, and degraded latency performance in high-density scenarios [4], [5], [30]. Although integrating UAVs as aerial base stations can alleviate these limitations by providing on-demand capacity and improved LoS conditions, existing approaches generally treat mobility management and resource allocation in isolation. Prior work predominantly relies on static, non-predictive, or loosely coupled models for either trajectory control [18], [19] or resource allocation and computation

offloading [2], [13], [14], [16], [17]. At the same time, handover mechanisms are typically handled using heuristic or threshold-based rules without predictive mobility awareness [1], [3], [20]–[22]. As a result, these fragmented designs fail to synchronize UAV motion with vehicle dynamics and network load, leading to inefficient channel usage and unstable service continuity during high-speed mobility. Motivated by these limitations, this work investigates the following core question: *How can a deterministic, jointly optimized protocol simultaneously govern UAV trajectory, radio resource allocation, and client handover to ensure ultra-low latency and robust service continuity in high-mobility UAV-RSU integrated vehicular networks?*

The main technical contributions of this paper are outlined as follows:

- 1) **Predictive Mobility Modeling:** We integrate a Kalman Filter-based mechanism to accurately predict the short-term position and velocity of high-mobility vehicles, enabling proactive resource planning and anticipatory, rather than reactive, handover decisions.
- 2) **Joint Deterministic Optimization:** We formulate and solve a novel joint optimization problem ($\mathcal{P}1$) that deterministically coordinates three critical variables—UAV trajectory, channel resource allocation, and client handover policy—to minimize overall service latency (T_{total}).
- 3) **Game-Theoretic Handover Protocol:** We propose a Cooperative Game Theory model wherein the RSU and the UAV are players. The handover decision is determined by the Nash Equilibrium derived from the players' utility functions (based on predicted connectivity and resource load), ensuring optimal resource utilization.
- 4) **Holistic Framework Validation:** We validate the end-to-end performance of the DCRT Optimization Framework using comprehensive simulations, demonstrating superior performance gains in both latency and throughput metrics compared to current non-predictive handover mechanisms.

The remainder of this paper is structured as follows: Section 2 critically reviews the related work in UAV-based V2X communication and resource management. Section 3 details the comprehensive system model and the underlying channel assumptions. Section 4 presents System Metrics and Optimization Variables. Section 5 presents the proposed Deterministic Coordinated Trajectory and Resource (DCRT) Optimization Framework, including the Kalman Filter-based prediction and the Cooperative Game Theory model. Section 6 describes the simulation setup and presents the performance evaluation results. Finally, Section 7 concludes the paper and discusses directions for future work.

TABLE 1: Representative Literature on UAV-Assisted Vehicular Networks: Objectives, Mechanisms, and Limitations

References	Main Objective	Key Mechanism / Methodology	Limitations Relative to This Work	Scope ($\mathcal{T}/\mathcal{R}/\mathcal{H}$)	Joint Opt.
[25]	Minimize delay in joint $\mathcal{T}$ - $\mathcal{R}$ - $\mathcal{H}$ design	Deterministic decomposition (matching/coalition; BCD/SCA) over $\mathcal{T}$, $\mathcal{R}$, $\mathcal{H}$	No KF-based prediction; no cooperative $\mathcal{H}$; $\mathcal{H}$ cost ignored; A_{avg} not reported	$\mathcal{T}, \mathcal{R}, \mathcal{H}$	×
[26]	Delay-energy reduction over $\mathcal{T}$ and $\mathcal{R}$	Federated multi-agent RL for joint $\mathcal{T}$ and $\mathcal{R}$ control	$\mathcal{H}$ implicit; no KF; non-deterministic; A_{avg} missing	$\mathcal{T}, \mathcal{R}$	×
[29]	Reduce AoI/cost under $\mathcal{T}$ - $\mathcal{R}$ - $\mathcal{H}$ coupling	SDN-based DRL for $\mathcal{T}$, $\mathcal{R}$, and $\mathcal{H}$ /offloading control	No deterministic $\mathcal{H}$; no KF; no A_{avg} ; weak system-level $\mathcal{T}/\mathcal{R}/\mathcal{H}$ coordination	$\mathcal{T}, \mathcal{R}, \mathcal{H}$	×
[27]	AoI + energy minimization via $\mathcal{T}/\mathcal{R}/\mathcal{H}$ decisions	SAC for $\mathcal{T}$ - $\mathcal{R}$ - $\mathcal{H}$ and offloading	No deterministic/cooperative $\mathcal{H}$; no KF; no area throughput metric	$\mathcal{T}, \mathcal{R}, \mathcal{H}$	×
[14]	Minimize offloading delay ($\mathcal{T}$ - $\mathcal{R}$)	Multi-agent SAC for trajectory $\mathcal{T}$ + resource $\mathcal{R}$ optimization	$\mathcal{H}$ not explicit; no KF; A_{avg} missing	$\mathcal{T}, \mathcal{R}$	×
[16]	Delay minimization w/ EH under $\mathcal{T}$ - $\mathcal{R}$ design	MADDPG for $\mathcal{T}$ and $\mathcal{R}$ decisions	$\mathcal{H}$ ignored; no KF; no area throughput metric	$\mathcal{T}, \mathcal{R}$	×
[2]	QoS-driven $\mathcal{R}$ + UAV placement ($\mathcal{T}$)	Mean-field RL for NOMA $\mathcal{R}$ and UAV $\mathcal{T}$ deployment	No $\mathcal{H}$; limited $\mathcal{T}$ coupling; no KF; no A_{avg}	$\mathcal{R}, \mathcal{T}$	×
[13]	System delay reduction via $\mathcal{T}/\mathcal{R}$	DDPG-LSTM for $\mathcal{T}$ and $\mathcal{R}$ with offloading	No $\mathcal{H}$; no KF; A_{avg} not evaluated	$\mathcal{T}, \mathcal{R}$	×
[1], [3], [20], [31]	Throughput improvement via $\mathcal{R}$ - $\mathcal{H}$ coordination	Deterministic LoS/SINR-based $\mathcal{R}$ and $\mathcal{H}$ rules	No game-theoretic $\mathcal{H}$; no $\mathcal{T}$ coupling; no A_{avg} metric	$\mathcal{R}, \mathcal{H}$	×
[17]	Minimize cost via $\mathcal{R}$ - $\mathcal{T}$ decisions	Distributed RL for $\mathcal{R}$ + heuristic UAV $\mathcal{T}$ deployment	$\mathcal{H}$ missing; no KF; no A_{avg} analysis	$\mathcal{R}, \mathcal{T}$	×
This Work (DCRT)	Joint minimization of latency + maximization of A_{avg} via $\mathcal{T}/\mathcal{R}/\mathcal{H}$	KF-based mobility prediction + Cooperative game-theoretic $\mathcal{H}$ + Deterministic joint $\mathcal{T}/\mathcal{R}/\mathcal{H}$ optimization	Provides anticipatory, deterministic RSU-UAV decisions with explicit $\mathcal{H}$ cost and URLLC constraints	$\mathcal{T}, \mathcal{R}, \mathcal{H}$	✓

Legend: $\mathcal{T}$ = UAV trajectory; $\mathcal{R}$ = resource allocation; $\mathcal{H}$ = handover. **Joint Opt.** = joint optimization of $\mathcal{T}/\mathcal{R}/\mathcal{H}$.

II. Related Work

The literature on UAV-assisted vehicular networks can be generally categorized into three primary research areas: (1) Resource Allocation (RA) in UAV Networks, (2) UAV Trajectory Optimization, and (3) Handover Mechanisms in Aerial-Ground Integrated Networks, as shown in Table 1 and 2.

A. Resource Allocation in UAV Networks

Early work in RA focused on static or semi-static deployment of UAVs to serve ground users [30]. Research in this domain typically involves maximizing the sum rate or minimizing transmit power by allocating sub-channels and power levels [24], [40]. For instance, studies such as [7] utilized convex optimization and game-theoretic formulations

to optimize power control subject to interference constraints. While effective for instantaneous resource management, these approaches are often computationally intensive and do not account for the rapid mobility inherent in VANETs. Furthermore, the decoupling of RA from the UAV's trajectory planning results in sub-optimal long-term performance, as the allocated resources quickly become outdated as vehicles move [1], [26], [28], [38], [39].

B. UAV Trajectory Optimization

Our research into UAV trajectory optimization aims to determine the optimal flight path (position in 3D space across time) to maximize connectivity or coverage time. Typical approaches involve DRL or sophisticated path-planning algorithms [8], [32], [36], [37], [50]. DRL-based

TABLE 2: Functional Categorization of UAV-Assisted Vehicular Network Literature

Category	Representative References
$\mathcal{T}$ only	[8], [23], [32]–[37]
$\mathcal{R}$ only	[1], [7], [24], [26], [28], [30], [38]–[40]
$\mathcal{H}$ only	[21], [22], [41]–[46]
$\mathcal{T} + \mathcal{R}$	[10], [47], [48]
$\mathcal{T} + \mathcal{H}$	[49]
$\mathcal{R} + \mathcal{H}$	[6], [13], [17], [25], [29]
$\mathcal{T} + \mathcal{R} + \mathcal{H}$	[25]

Legend: $\mathcal{T}$ = UAV trajectory; $\mathcal{R}$ = resource allocation; $\mathcal{H}$ = handover; $\mathcal{T} + \mathcal{R}$ = joint trajectory + resource allocation; $\mathcal{T} + \mathcal{H}$ = joint trajectory + handover; $\mathcal{R} + \mathcal{H}$ = joint resource allocation + handover; $\mathcal{T} + \mathcal{R} + \mathcal{H}$ = joint trajectory + resource allocation + handover.

methods, while adaptive, often lack the guaranteed convergence and interpretability required for safety-critical ITS applications. Non-DRL methods often treat vehicle positions as fixed or use simple linear motion models [23], [33], [34]. These approaches fail to accurately predict rapid state changes (acceleration/deceleration), making them unsuitable for deterministic, ultra-low-latency requirements. Critically, most trajectory work assumes an existing resource allocation policy rather than jointly optimizing the two [35].

C. Handover Mechanisms in Aerial-Ground Integrated Networks

Handover protocols are essential for maintaining service continuity as vehicles move between RSU coverage and UAV coverage. Traditional threshold-based methods (e.g., those triggered solely by Received Signal Strength Indicator (RSSI) dropping below a fixed level) [21] are reactive and suffer from the ping-pong effect in dynamic environments. More advanced approaches, such as those using Markov decision processes (MDP) [41], [43], offer improved decision-making but often require extensive training data and are computationally expensive for real-time operation. Furthermore, existing literature often treats handover as a single event rather than a structured multi-stage process. As detailed in the architecture proposed in [51], a robust handover procedure in Space-Air-Ground Integrated Networks (SAGIN) consists of three distinct phases: handover trigger, handover execution, and handover completion. While [51] emphasizes a selection scheme based on multi-attribute quality of service (QoS) and network status, our DCRT framework extends this by making the trigger phase predictive via Kalman filtering and the execution phase deterministic through game-theoretic coordination. The primary limitation across all existing handover literature [22], [42], [44]–[46] is the lack of an integrated predictive state machine (such as a Kalman Filter) combined with a game-theoretic decision model that accounts for the resource load of both the source and

target base stations (RSU and UAV) simultaneously before executing the handover.

D. Summary and Research Critique

Beyond trajectory and handover optimization, recent literature has explored diverse dimensions of resource coordination in vehicular networks. For instance, the RAISE framework [52] investigates the strategic placement of Reconfigurable Intelligent Surfaces (RIS) to maximize task throughput in multi-server vehicular edge computing by mitigating ground-level obstructions. Complementarily, the Prioritized Information Bottleneck (PIB) framework [53] addresses the challenge of data redundancy in edge video analytics by using distributed online learning to prioritize essential information. These studies highlight the importance of joint physical-layer and application-layer optimization, providing a broader context for the cooperative resource management proposed in our DCRT framework. The limitations of the existing literature underscore the need for our integrated approach. Table 3 summarizes the key limitations of current approaches and highlights the unique contribution of the DCRT framework. List of key notations used are listed in Table 4.

III. System Model and Assumptions

A. The Handover Procedure in DCRT

To ensure seamless transitions and ultra-low latency, the DCRT framework formally defines the handover procedure into three sequential phases, following the conceptual model in [51] but optimized for high-mobility vehicular environments:

- 1) **Handover Trigger Phase:** In DCRT, the trigger is not solely based on instantaneous RSSI. Instead, it is initiated when the Kalman Filter predicts that the current link quality will degrade below a threshold $SINR_{th}$ within the next ΔT slots, or when the game-theoretic utility U_{sys} indicates that a neighbor node (UAV or RSU) can provide a superior Pareto-optimal resource state.
- 2) **Handover Execution Phase:** Once triggered, the RSU and UAV engage in the Cooperative Game. The execution involves the deterministic assignment of sub-channels and the synchronization of the UAV trajectory to intercept the vehicle's predicted path. This minimizes the interruption time by pre-allocating resources at the target node before the connection is switched.
- 3) **Handover Completion Phase:** The completion phase is defined by the successful acknowledgment of the first packet on the new link and the update of the global state $h_k^i[n]$. Unlike traditional SAGIN procedures that may involve high signaling overhead [51], DCRT utilizes the shared backbone between the RSU and the UAV controller to update the state deterministically, eliminating the need for extensive back-and-forth control signaling.

TABLE 3: Comparison of Existing and Proposed Frameworks

Feature	UAV-Centric RA X1	Trajectory Optimization (DRL/Heuristic) X2,X3	Threshold/MDP Handover X4	DCRT Optimization Framework (Proposed)
Prediction Model	None/Simple Linear	Simple (often non-deterministic)	Simple/Stochastic (MDP)	Kalman Filter (Deterministic State)
Optimization Focus	Resource Allocation only	Trajectory/Coverage only	Handover Latency only	Joint Optimization ($\mathcal{P}1$)
Core Mechanism	Convex Optimization	DRL/Path Planning	RSSI/MDP	Cooperative Game Theory
Decision Scope	Instantaneous	Short-Term Trajectory	Reactive	Anticipatory and Coordinated
Achieves Joint Optimization	×	×	×	✓ (Trajectory, RA, Handover)

TABLE 4: Summary of Key Notations

Symbol	Description
K	Total number of vehicles in the network segment
M	Number of available Road Side Units (RSUs)
U	Number of active Unmanned Aerial Vehicles (UAVs)
$\mathcal{H}_k$	Handover association state for vehicle k , $\mathcal{H}_k \in \{0, 1, \dots, M + U\}$
$\gamma_{k,n}$	Signal-to-Interference-plus-Noise Ratio (SINR) for vehicle k to node n
B_k	Bandwidth allocated to vehicle k
$\mathcal{P}_k$	Transmission power of vehicle k
σ^2	Additive White Gaussian Noise (AWGN) power
$d_{R,n}$	Euclidean distance between vehicle R and node n
α	Path loss exponent
$\Phi(\cdot)$	Potential function of the coordination game

To clarify the operational principles of the proposed framework, we define determinism in the context of DCRT as the property whereby the handover decisions and resource states are derived through a sequence of mathematical operations, rather than stochastic approximation or random-access procedures. Specifically, the framework is deterministic in three dimensions:

- **State Estimation:** The use of a closed-form Kalman Filter (KF) ensures that for a given set of previous observations, the predicted trajectory window is uniquely determined.
- **Decision Logic:** Unlike Reinforcement Learning (RL) approaches that may rely on exploration or probabilistic policies, DCRT utilizes an exact best-response dynamic.
- **Convergence:** The game-theoretic coordination is governed by a monotonic potential function, ensuring that the system always converges to a stable Nash Equilibrium (NE) from any initial state without the variance associated with meta-heuristic searches.

For the system model, we consider a hybrid access network deployed along a straight highway segment. The infrastructure consists of a single fixed RSU, a rotary-wing UAV acting as an aerial base station, and a set of vehicular users

$\mathcal{K} = \{1, 2, \dots, K\}$. The system operates over a discrete time horizon of N equal-length time slots, $\mathcal{N} = \{1, 2, \dots, N\}$, where each slot has duration τ as shown in Fig. 1.

B. Network Geometry and Mobility

The RSU is static and located at a known position $\mathbf{L}_R = (x_R, y_R, 0)$. The position of vehicle k in the time slot n is denoted by

$$\mathbf{v}_k[n] = (x_k[n], y_k[n], 0), \quad k \in \mathcal{K}, n \in \mathcal{N}, \quad (1)$$

and the vehicles are assumed to move predominantly along the x -axis following lane-dependent longitudinal speeds. A standard constant-velocity model is adopted for the nominal state transition:

$$x_k[n+1] = x_k[n] + v_k\tau + w_x[n], \quad (2)$$

$$y_k[n+1] = y_k[n] + w_y[n], \quad (3)$$

where v_k is derived from a lane-dependent distribution. To ensure robustness against abrupt maneuvers such as sudden braking or lane changes, which violate the linear constant-velocity assumption, we model these deviations as stochastic process noise $w[n] \sim \mathcal{N}(0, Q)$. The Kalman Filter's recursive update step utilizes real-time position measurements to adjust the state estimate, effectively mitigating the impact

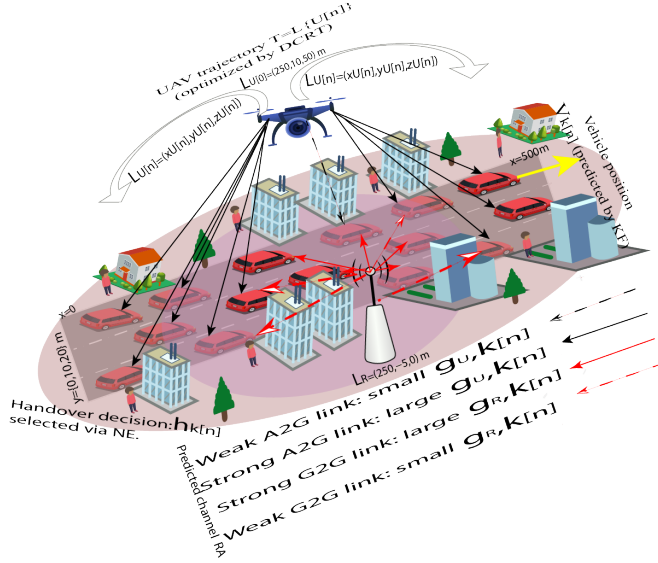


FIGURE 1: System model of the proposed RSU-UAV integrated vehicular network. Vehicles move along the road with states $\mathbf{v}_k[n]$ and predicted positions $\hat{\mathbf{v}}_k[n+m]$ obtained via Kalman filtering. The UAV follows a dynamic trajectory $\mathcal{T} = \{\mathbf{L}_U[n]\}$ optimized by the DCRT framework. Predicted A2G and G2G channel gains ($g_{U,k}, g_{R,k}$) determine resource allocation $R_k^i[n]$ and the cooperative handover decision $h_k[n]$. Solid black/red arrows represent strong UAV/RSU links; dashed arrows indicate weak links.

of non-linear accelerations and maintaining predictive accuracy. The UAV provides on-demand coverage and mobility support. Its trajectory is described by the position vector

$$\mathbf{L}_U[n] = (x_U[n], y_U[n], z_U[n]), \quad n \in \mathcal{N}, \quad (4)$$

where the altitude is constrained by

$$H_{\min} \leq z_U[n] \leq H_{\max}, \quad \forall n \in \mathcal{N}. \quad (5)$$

Between consecutive slots, the UAV speed is limited according to

$$V_{\min} \leq \frac{\|\mathbf{L}_U[n] - \mathbf{L}_U[n-1]\|}{\tau} \leq V_{\max}, \quad \forall n \in \mathcal{N}, \quad (6)$$

which captures both mechanical and regulatory constraints on UAV motion.

C. Stability and Convergence

To ensure the stability and convergence of our handover decisions, we model the interaction between the RSU and UAV as a coordination game. Let $\mathcal{A}_k$ be the set of possible association actions for vehicle k . The system utility is governed by a potential game structure, which guarantees the existence of at least one pure-strategy Nash Equilibrium (NE). The proposed coordination game is a discrete Exact Potential Game (EPG).

Proof:

Consider a potential function $\Phi(\mathcal{H})$ defined as the aggregate system utility across all vehicles:

$$\Phi(\mathcal{H}) = \sum_{k \in \mathcal{K}} \sum_{i \in \{R, U\}} h_k^i U_k^i(\mathcal{T}, \mathcal{R}) \quad (7)$$

where U_k^i is the utility of vehicle k associated with node i . When a single vehicle k switches its association from node i to node j (i.e., a handover occurs), the change in its individual utility $\Delta u_k = U_k^j - U_k^i$. Under our deterministic coordination, since the resources at nodes i and j are recalculated based on the new association state to satisfy $\mathcal{P}1$, the change in the global potential function Φ is exactly equal to Δu_k :

$$\Phi(h_k^j, \mathcal{H}_{-k}) - \Phi(h_k^i, \mathcal{H}_{-k}) = U_k^j - U_k^i \quad (8)$$

This satisfies the definition of an exact potential game. Because the strategy space (association states) is finite and Φ is bounded, the sequence of best-response updates must terminate at a local maximum of Φ , which corresponds to a pure-strategy Nash Equilibrium. ■

Existence and Uniqueness: While every potential game possesses at least one NE, uniqueness is not guaranteed in non-convex multi-user environments. However, in the DCRT framework, uniqueness is practically enforced by the deterministic tie-breaking rule in the best-response dynamic: if multiple association states yield identical utility, the state that minimizes the physical distance (and thus maximizes LoS probability) is chosen. This ensures that the system settles into a single, reproducible NE.

Avoiding Local Optima: To mitigate the risk of converging to sub-optimal local maxima of Φ , DCRT employs a multi-start deterministic search during the initialization of each time slot. By seeding the best-response algorithm with both the previous state (for continuity) and a predicted optimal state (derived from the Kalman Filter's trajectory projection), the framework significantly increases the probability of reaching the global optimum. Furthermore, the inclusion of the trajectory variable $\mathcal{T}$ in the utility function allows the UAV to move proactively, effectively stretching the potential landscape to escape shallow local optima that would otherwise trap a static infrastructure network.

D. Channel Modelling

Let $h_{i,k}[n]$ denote the complex channel coefficient between node $i \in \{R, U\}$ and vehicle k in slot n . We distinguish between two types of links: the Ground-to-Ground (G-to-G) RSU-vehicle channel and the Air-to-Ground (A-to-G) UAV-vehicle channel.

1) RSU-Vehicle (G-to-G) Channel

The RSU-vehicle link mainly experiences non-line-of-sight (NLoS) propagation, dominated by blockages from buildings and other vehicles. Let $d_{R,k}[n] = \|\mathbf{L}_R - \mathbf{v}_k[n]\|$ be the Euclidean distance between the RSU and vehicle k in slot

n . The large-scale channel gain is modelled as

$$g_{R,k}[n] = L_0 \left(\frac{d_{R,k}[n]}{d_{\text{ref}}} \right)^{-\alpha_R}, \quad (9)$$

where L_0 is the channel gain at a reference distance d_{ref} and $\alpha_R > 2$ is the path-loss exponent for the RSU link. Small-scale fading is modelled as Rayleigh, so that the resulting instantaneous channel power gain $h_{R,k}[n] \sim \mathcal{CN}(0, g_{R,k}[n])$.

2) UAV–Vehicle (A–to–G) Channel

The UAV–vehicle channel benefits from an elevated vantage point and, hence, a higher probability of line-of-sight (LoS) connectivity. Let $d_{U,k}[n] = \|\mathbf{L}_U[n] - \mathbf{v}_k[n]\|$ and $\theta_{U,k}[n]$ be the corresponding elevation angle. The LoS probability is modelled as

$$P_{\text{LoS}}[n] = \frac{1}{1 + a \exp(-b(\theta_{U,k}[n] - a))}, \quad (10)$$

where a and b are environment-dependent constants.

The average large-scale channel gain between the UAV and vehicle k is then expressed as

$$g_{U,k}[n] = P_{\text{LoS}}[n] g_{\text{LoS}}[n] + (1 - P_{\text{LoS}}[n]) g_{\text{NLoS}}[n], \quad (11)$$

where $g_{\text{LoS}}[n]$ and $g_{\text{NLoS}}[n]$ denote the LoS and NLoS gains, respectively, each following a distance-dependent power law with appropriate exponents. Small-scale fading is typically modelled as Rician for LoS and Rayleigh for NLoS conditions. The resulting instantaneous channel power gain is $|h_{U,k}[n]|^2$ with mean $g_{U,k}[n]$.

E. SINR and Link Capacity

Both RSU and UAV operate over a shared band divided into C orthogonal channels. If vehicle k transmits in slot n on channel c , with transmit power P_T , the received power at node $i \in \{R, U\}$ is

$$P_{i,k,c}[n] = P_T |h_{i,k,c}[n]|^2. \quad (12)$$

Let $\mathcal{T}_c[n]$ denote the set of vehicles transmitting on channel c in slot n . The instantaneous SINR for vehicle k at node i on that channel is

$$\gamma_{i,k,c}[n] = \frac{P_{i,k,c}[n]}{N_0 + \sum_{j \in \mathcal{T}_c[n], j \neq k} P_{i,j,c}[n]}, \quad (13)$$

where N_0 is the noise power. To account for the impact of correlated interference in dense vehicular environments, we model the aggregate interference term in (10) as a spatially-correlated stochastic process. A packet from vehicle k is considered successfully decoded if the best available link satisfies the SINR criteria, i.e., $\max\{\gamma_{R,k,c}[n], \gamma_{U,k,c}[n]\} \geq \gamma_{th}$. However, in scenarios where both links are simultaneously marginal (i.e., $\gamma_{i,k,c} \approx \gamma_{th}$), the proposed DCRT framework treats the resulting decoding uncertainty as a reduction in the predicted utility U_{sys} . This triggers the cooperative game-theoretic module to either initiate a proactive handover to a less congested channel or adjust the UAV trajectory to

improve the Air-to-Ground (A2G) link budget, thereby mitigating the risk of simultaneous link failure. The achievable rate on a successfully decoded link is then expressed as $R_k[n] = B \log_2(1 + \gamma_{i,k,c}[n])$, where B is the channel bandwidth.

F. QoS Metric Definition

The Quality of Service (QoS) for each vehicle k , denoted as $\Phi_k[n]$, is defined as a normalized multi-objective satisfaction index. It quantifies the degree to which the current network state meets the protocol requirements:

$$\Phi_k[n] = \exp\left(-\alpha \frac{T_{\text{total},k}}{\bar{T}}\right) \cdot \left(\frac{R_k[n]}{R_{\text{req}}}\right) \quad (14)$$

where $T_{\text{total},k}$ is the predicted service delay, $R_k[n]$ is the achievable data rate, and R_{req} is the minimum rate requirement. The exponential decay constant α ensures that the utility drops sharply if the latency exceeds the threshold $\bar{T}$.

IV. System Metrics and Optimization Variables

We evaluate the network performance using two primary metrics:

- 1) **Total service time** $T_{\text{total},k}$: the number of time slots from the instant the vehicle k requests service until all its data has been successfully delivered.
- 2) **Average area throughput** A_{avg} : the long-term average sum rate per unit area of the highway segment.

The proposed Deterministic Coordinated Trajectory and Resource (DCRT) framework jointly optimizes three sets of variables:

- 1) **UAV trajectory** ($\mathcal{T}$): the sequence of UAV positions over the time horizon of N slots,
- 2) **Resource allocation** ($\mathcal{R}$): the assignment of channel bandwidth (or sub-channels) to vehicles by the serving base station (RSU or UAV), which specifies the rate (or bandwidth) allocated to each vehicle in each time slot.
- 3) **Handover policy** ($\mathcal{H}$): the collection of binary association variables,

$$\mathcal{T} = \{\mathbf{L}_U[n]\}_{n=1}^N, \quad (15)$$

where $\mathbf{L}_U[n]$ denotes the UAV position in slot n .

$$\mathcal{H} = \{h_k^i[n]\}, \quad (16)$$

where $h_k^U[n] = 1$ indicates that vehicle k is associated with the UAV in slot n , $h_k^R[n] = 1$ indicates RSU association, and the exclusivity constraint

$$h_k^U[n] + h_k^R[n] = 1 \quad (17)$$

holds for all k and n .

The core objective of the DCRT framework is to formulate and solve a unified joint optimization problem that simultaneously minimizes network latency and maximizes system throughput, while respecting the mobility, association, and

resource constraints imposed by the UAV-RSU integrated architecture. To balance the competing requirements of service delay and spectral efficiency, we define the joint objective function in a dimensionally consistent form. Let $\bar{T}$ denote a normalization constant for the maximum allowable service time and $\bar{A}$ denote the target system throughput. The revised objective function for problem $\mathcal{P}1$ is:

$$\min_{\mathcal{T}, \mathcal{H}, \mathcal{C}} \quad \omega_L \left(\frac{T_{total,k}}{\bar{T}} \right) - \omega_A \left(\frac{A_{avg}}{\bar{A}} \right) \quad (18)$$

where both terms are now dimensionless. This formulation follows the principles of Multi-Attribute Utility Theory, where heterogeneous metrics are mapped to a common scale $[0, 1]$ before aggregation. This joint design naturally leads to the following mixed-integer, non-convex optimization problem:

$$\mathcal{P}1: \min_{\mathcal{T}, \mathcal{R}, \mathcal{H}} \quad \omega_L \sum_{k=1}^K T_{total,k} - \omega_A A_{avg} \quad (19a)$$

$$\text{s.t. C1: } V_{\min} \leq \frac{\|\mathbf{L}_U[n] - \mathbf{L}_U[n-1]\|}{\tau} \leq V_{\max}, \quad \forall n, \quad (19b)$$

$$\text{C2: } H_{\min} \leq z_U[n] \leq H_{\max}, \quad \forall n, \quad (19c)$$

$$\text{C3: } \sum_{k \in \mathcal{K}} R_k^i[n] \leq B_i[n], \quad i \in \{R, U\}, \quad \forall n, \quad (19d)$$

$$\text{C4: } h_k^R[n] + h_k^U[n] = 1, \quad h_k^i[n] \in \{0, 1\}, \quad \forall k \in \mathcal{K}, \quad \forall n, \quad (19e)$$

$$\text{C5: } T_{total,k} \leq T_{\max}, \quad \forall k \in \mathcal{K}, \quad (19f)$$

where $R_k^i[n]$ denotes the rate allocated to vehicle k by node $i \in \{R, U\}$, $B_i[n]$ represents the available bandwidth at node i in slot n , and $\omega_L, \omega_A > 0$ are weighting factors that balance the conflicting objectives of minimizing latency and maximizing throughput. Problem $\mathcal{P}1$ therefore captures the inherent coupling between user mobility, base-station association, and spectral resource allocation. The solution approach for this mixed-integer, non-convex problem is presented in the section below. The weighting factors ω_L and ω_A are chosen based on the tolerable latency levels and traffic conditions. For safety-critical ITS applications where latency is the primary constraint, a higher ω_L is assigned to ensure minimum service completion time $T_{total,k}$. In contrast, for data-intensive and time tolerant applications under high vehicle density, ω_A is increased to prioritize spectral efficiency and area throughput. Our sensitivity analysis (discussed further in Sec. VI) indicates that the DCRT framework maintains a performance margin of at least over conventional protocols across a wide range of weight ratios ($\omega_L/\omega_A \in [0.1, 10]$), confirming that the gains are driven by the deterministic coordination of trajectory and handover rather than the specific tuning of weights.

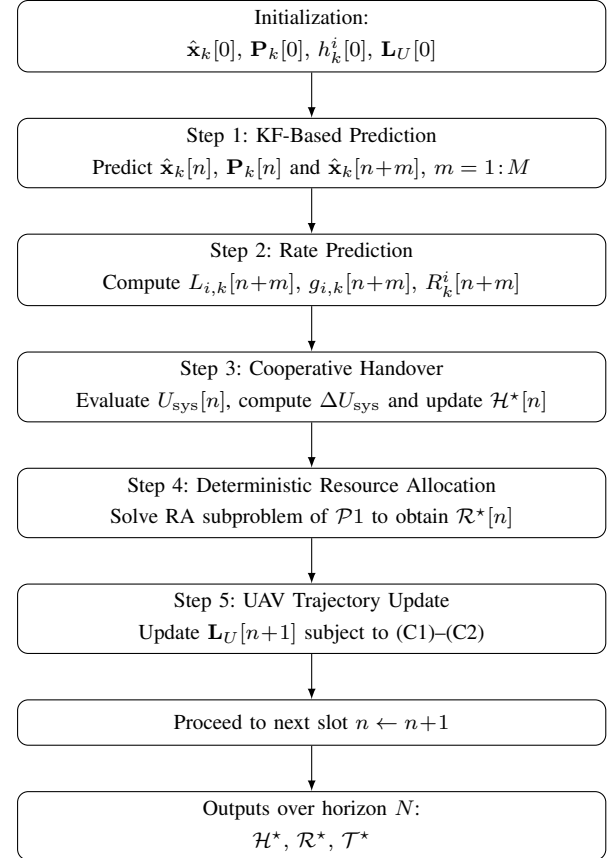


FIGURE 2: Flowchart of the Deterministic Coordinated Trajectory and Resource (DCRT) framework, integrating KF-based predictive mobility, cooperative handover, deterministic resource allocation, and UAV trajectory update to solve $\mathcal{P}1$.

V. The Deterministic Coordinated Trajectory and Resource (DCRT) Framework

Directly solving $\mathcal{P}1$ is computationally prohibitive due to the mixed-integer structure and the non-convex coupling between UAV trajectory, handover, and rate allocation. The DCRT framework addresses this difficulty by decomposing $\mathcal{P}1$ into two interacting modules: (i) predictive vehicle state estimation and (ii) cooperative game-theoretic handover and resource allocation. The resulting decisions are deterministic functions of the predicted states, which guarantees reproducible behaviour, high accuracy, and reliability.

A. Predictive Vehicle State Estimation via Kalman Filter

Accurate prediction of vehicle motion is essential for anticipatory resource allocation and UAV trajectory planning. To obtain deterministic short-term mobility forecasts, we employ a Kalman Filter (KF), which serves as the optimal state estimator for linear dynamic systems subject to Gaussian process and measurement noise. For each vehicle k , the KF predicts its position and velocity for the next M time slots.

1) State Vector Definition

For vehicle k in slot n , the state vector is defined as

$$\mathbf{x}_k[n] = [x_k[n], y_k[n], v_{x,k}[n], v_{y,k}[n]]^T, \quad (20)$$

where $(x_k[n], y_k[n])$ denote the planar coordinates, and $(v_{x,k}[n], v_{y,k}[n])$ represent the corresponding velocity components along the horizontal axes.

2) KF Process and Measurement Models

The KF operates in two stages: a *prediction step*, which propagates the state forward using vehicle dynamics and an *update step* that refines this prediction based on newly observed measurements.

a: Prediction Step (Time Update)

The a priori estimate of the state and its covariance at slot n are given by

$$\hat{\mathbf{x}}_k^-[n] = \mathbf{F} \hat{\mathbf{x}}_k[n-1] + \mathbf{u}[n-1], \quad (21)$$

$$\mathbf{P}_k^-[n] = \mathbf{F} \mathbf{P}_k[n-1] \mathbf{F}^T + \mathbf{Q}, \quad (22)$$

where $\hat{\mathbf{x}}_k[n-1]$ is the posterior estimate at slot $n-1$, $\mathbf{F}$ is the state-transition matrix (e.g., constant-velocity model), $\mathbf{u}[n-1]$ is an optional control input, $\mathbf{P}_k[n-1]$ is the estimation error covariance, and $\mathbf{Q}$ is the process noise covariance.

b: Update Step (Measurement Update)

Given a noisy position measurement $\mathbf{z}_k[n]$ and the measurement matrix $\mathbf{H}$, the measurement update is performed as

$$\mathbf{K}[n] = \mathbf{P}_k^-[n] \mathbf{H}^T (\mathbf{H} \mathbf{P}_k^-[n] \mathbf{H}^T + \mathbf{R})^{-1}, \quad (23)$$

$$\hat{\mathbf{x}}_k[n] = \hat{\mathbf{x}}_k^-[n] + \mathbf{K}[n] (\mathbf{z}_k[n] - \mathbf{H} \hat{\mathbf{x}}_k^-[n]), \quad (24)$$

$$\mathbf{P}_k[n] = (\mathbf{I} - \mathbf{K}[n] \mathbf{H}) \mathbf{P}_k^-[n], \quad (25)$$

where $\mathbf{K}[n]$ is the Kalman gain and $\mathbf{R}$ is the measurement noise covariance.

By iterating the prediction step for $m = 1, \dots, M$, the filter produces the future predicted states $\hat{\mathbf{x}}_k[n+m]$. These short-term forecasts of vehicle position and velocity are then used to estimate future channel conditions and serve as the input to the game-theoretic decision module.

B. Cooperative Game Theory for Handover and Resource Allocation

Given the predicted vehicle states over the horizon M , the RSU and the UAV jointly determine (i) which vehicles they will serve and (ii) how their respective spectral resources are allocated. This interaction is captured using a cooperative game denoted by $\mathcal{G} = (\mathcal{I}, \mathcal{S}, \mathcal{U})$.

1) Players, Strategies, and Utility

a: Players ($\mathcal{I}$):

The game involves two decision-making entities: the RSU (player R) and the UAV (player U).

b: Strategies ($\mathcal{S}$):

The strategy space corresponds to the handover and association decisions for all vehicles. Specifically,

$$\mathcal{S} = \{\mathcal{H} = \{h_k^i[n] : i \in \{R, U\}, k \in \mathcal{K}\}, \quad (26)$$

where $h_k^U[n] = 1$ denotes that vehicle k is associated with the UAV in slot n , $h_k^R[n] = 1$ denotes RSU association, and the exclusivity condition

$$h_k^U[n] + h_k^R[n] = 1 \quad (27)$$

holds for all k and n .

c: Utility ($\mathcal{U}$):

The global system utility U_{sys} in Eq. (24) is the aggregate of these individual QoS metrics minus a load imbalance penalty. Specifically:

$$U_{sys} = \sum_{k \in \mathcal{K}} \Phi_k[n] - \lambda \cdot \text{Var}(\mathbf{L}) \quad (28)$$

where $\text{Var}(\mathbf{L})$ is the variance of the load across the RSU and UAV. This quantitative formulation ensures that the handover mechanism prioritizes vehicles with the highest potential QoS gain while preventing any single node from becoming a bottleneck.

C. Predictive Trajectory Update

While the handover logic determines vehicle association, the UAV trajectory update ensures the aerial base station remains at the most optimal position to serve its assigned cluster $\mathcal{K}_U[n]$. Given the predicted states of the assigned vehicles $\{\hat{\mathbf{x}}_k[n+1]\}_{k \in \mathcal{K}_U[n]}$ from the KF, the UAV's target location $\mathbf{L}_U[n+1] = [x_U, y_U, H]$ is calculated by solving the following sub-problem:

$$\mathbf{L}_U[n+1] = \arg \max_{\mathbf{L}} \sum_{k \in \mathcal{K}_U[n]} w_k \log_2(1 + \gamma U, k(\mathbf{L})) \quad (29)$$

subject to the physical displacement constraint:

$$|\mathbf{L}_U[n+1] - \mathbf{L}_U[n]| \leq V_{max} \cdot \tau \quad (30)$$

where V_{max} is the maximum UAV speed and τ is the slot duration. In our DCRT implementation, we simplify this to a weighted centroid of the predicted vehicle positions, which offers a closed-form solution with complexity $O(|\mathcal{K}_U|)$, ensuring real-time feasibility for high-mobility scenarios.

D. Inter-module Coupling

The interaction between the predictive and game-theoretic modules is governed by a feedback loop where the estimator updates the transition probability P , which is then used by the game module to refine the Nash equilibrium strategy π^* . The algorithm governing this mechanism is illustrated below:

Algorithm 1 Iterative Interaction between Predictive and Game Modules

```

1: Initialize: State  $s_0$ , Prediction Horizon  $H$ , Convergence threshold  $\epsilon$ 
2: while Task not complete do
3:   Step 1: Predictive Estimation
4:   Generate predicted trajectory  $\hat{X}$  based on current observations  $s_t$ .
5:   Step 2: Game-Theoretic Optimization
6:   Input  $\hat{X}$  into module  $\mathcal{G}$  to solve for Nash Equilibrium strategy  $\pi^*$ .
7:   Step 3: Interaction & Feedback
8:   if Strategy  $\pi^*$  significantly alters environment state then
9:     Re-estimate  $\hat{X}$  conditioned on ego-action  $a \in \pi^*$ .
10:   end if
11:   Step 4: Execution
12:   Apply first action  $a_t$  from  $\pi^*$ ; observe new state  $s_{t+1}$ .
13: end while

```

1) Optimal Handover Determination

The optimal handover policy $\mathcal{H}^*$ is obtained through an iterative best-response procedure that seeks the cooperative Nash Equilibrium (NE) of the RSU-UAV game. At the NE, neither the RSU nor the UAV can unilaterally improve the system utility by altering the service decision of any vehicle, and thus the resulting policy maximizes U_{sys} under the predicted mobility and resource constraints.

The decision process evaluates, for every vehicle k and each time slot n , the effect of switching its association between the RSU and the UAV. The incremental improvement in system utility is computed as

$$\Delta U_{\text{sys}} = U_{\text{sys}}^{(\text{switch})} - U_{\text{sys}}^{(\text{current})}, \quad (31)$$

where $U_{\text{sys}}^{(\text{switch})}$ denotes the system utility obtained after hypothetically changing $h_k^i[n]$, and $U_{\text{sys}}^{(\text{current})}$ corresponds to the current association. A handover from node i to node i' is accepted only when $\Delta U_{\text{sys}} > \epsilon$, with $\epsilon > 0$ acting as a tolerance to prevent unnecessary or marginally beneficial handovers. The iterative best-response dynamic continues until the marginal increase in the potential function, U_{sys} , falls below a threshold ϵ . Specifically, we set the handover threshold to 10^{-3} . Since the system utility is normalized (as defined in Eq. 15a), U_{sys} typically ranges between 0 and 1. A threshold of 10^{-3} therefore represents a 0.1% improvement in total system satisfaction. This value is chosen to ensure a high-precision Nash Equilibrium while maintaining a low computational footprint, consistent with established convergence criteria for potential games in wireless resource management [54].

The algorithm iteratively checks the service assignment for each vehicle k over the prediction window M :

- 1) **Initialization:** Begin with the current association map $\mathcal{H}[n]$ generated using predicted vehicle states over the horizon M .
- 2) **Iterative Optimization:** For each time slot n , iterate through all vehicles k and calculate the utility change ΔU_{sys} if the vehicle is handed over (i.e., changing $h_k^i[n]$).
- 3) **Decision:** The handover $k \rightarrow i'$ is executed only if $\Delta U_{\text{sys}} > \epsilon$, where ϵ is a small threshold, ensuring the change benefits the overall system and moves towards the NE.

The game-theoretic handover model can be formally classified as a Potential Game. In such games:

Convergence: The iterative best-response (IBR) procedure is guaranteed to converge to a Nash Equilibrium (NE) because each successful move (where $\Delta U_{\text{sys}} > \epsilon$) increases a global potential function (in this case, the system utility U_{sys}). Since the strategy space (association of K vehicles) is finite and the utility is bounded, the algorithm must reach a steady state in a finite number of steps.

Uniqueness: While potential games may have multiple NEs, the use of a deterministic initialization (starting from the current association $\mathcal{H}[n]$) and positive tolerance ϵ ensures that the system consistently reaches a stable and high-utility equilibrium.

Rapidly Changing States: The rapid changes are handled by the Kalman Filter (KF), which provides the game with a short-term stable window (M slots). By the time the state changes significantly, the game has already converged for the current slot's snapshot, and the trajectory update then physically moves the UAV to maintain the optimality of that NE. This game-theoretic approach directly addresses the conflicting objectives of the base stations. It provides a deterministic method to decide the optimal resource allocation $\mathcal{R}$ and handover policy $\mathcal{H}$, which in turn dictates the optimal UAV trajectory $\mathcal{T}$ (as the UAV is programmed to move to the best predicted location to satisfy the new service assignments).

E. Orthogonal Channel Allocation and Bandwidth Splitting

Given the set of vehicles associated with the RSU ($\mathcal{K}_R$) and the UAV ($\mathcal{K}_U$), the total available bandwidth B is divided into C orthogonal sub-channels to eliminate co-tier interference. Each vehicle k is allocated a bandwidth $b_k = B/C$. The resource allocation sub-problem optimizes the number of channels $|\mathcal{C}_i|$ assigned to node $i \in \{R, U\}$ to satisfy the predicted demand:

$$\min_{\mathcal{C}_R, \mathcal{C}_U} \sum_{i \in R, U} i \sum_{k \in \mathcal{K}_i} k (Req - \hat{R}_k[n+1])^2 \quad (32)$$

subject to:

$$|\mathcal{C}_R| + |\mathcal{C}_U| \leq C, \quad P_k \leq P_{\text{max}} \quad (33)$$

where $\hat{R}_k[n+1]$ is the rate calculated using the KF-predicted distance. This allocation is dynamic: if the KF predicts

a cluster of vehicles moving into the UAV's coverage, the DCRT framework proactively shifts sub-channels from the RSU to the UAV in the next time slot, ensuring that bandwidth follows the predicted traffic density.

F. Computational Complexity Analysis

The efficiency of the DCRT framework is due to its deterministic nature, avoiding high-dimensional exhaustive searches. The total complexity per time slot n is the sum of its four functional modules:

State Prediction (KF): For K vehicles, the Kalman Filter update involves matrix multiplications of constant dimensions (state vector size $m = 4$). Thus, the complexity is $O(K)$.

Handover Coordination (Game): As discussed in Sec. V-B, the potential game uses iterative best-response. Given I iterations to reach a Nash Equilibrium and two candidate nodes (RSU/UAV), the complexity is $O(I \cdot K)$. Empirically, $I \leq 5$, making it near-linear.

Trajectory Update: The weighted centroid calculation for the UAV's next location based on K_U associated vehicles is $O(K_U)$.

Resource Allocation: The bandwidth splitting sub-problem involves a ratio-based allocation or a priority queue for C channels, resulting in $O(C \log C)$ or $O(K)$.

Total Complexity: The overall per-slot complexity is $O(I \cdot K + C \log C)$. Since I and C are small constants in a single highway segment, the DCRT framework scales linearly with the number of vehicles, ensuring real-time execution in ultra-low latency scenarios.

VI. Performance Evaluation and Results

This section describes the simulation environment, the parametrization of the RAC, PAS, and DCRT schemes, and the numerical results that demonstrate the latency and throughput gains achieved by the proposed DCRT optimization framework.

A. Simulation Setup and Parameters

The DCRT framework is evaluated using a time-slotted simulation environment modeling a 500 m straight highway segment. A single RSU is located near the center, and one UAV provides auxiliary coverage. Vehicle mobility is simulated using a realistic vehicular mobility model (e.g., constant acceleration/deceleration combined with random noise, which is effectively mitigated by the Kalman Filter). While the baseline simulation utilizes a 500 m straight highway segment with a centrally located RSU to establish a canonical performance benchmark, the DCRT framework is adaptive to highway geometry. The underlying Kalman Filter-based prediction and the cooperative game-theoretic handover logic utilize instantaneous and predicted Euclidean distances between nodes, which are calculated independently of the road's curvature. In scenarios where highway geometry deviates from a linear path—or where RSU placement is asymmetrical, the joint optimization problem (P1) enables

the UAV to dynamically adjust its 3D trajectory $\mathcal{T}$ to balance the geometric coverage gaps. This ensures that the reported gains in latency (61%) and throughput (34.6%) are representative of the proposed work's flexibility to adapt to non-idealized infrastructure deployments. Key simulation parameters are summarized in Table 5.

B. Baseline Comparison Protocols

The DCRT framework is compared against two conventional protocols representing the state-of-the-art in non-game-theoretic handover systems:

- 1) **Reactive RSS-Threshold Handover (RAC, Baseline 1):** This is a purely reactive protocol where a vehicle initiates handover only when the Received Signal Strength Indicator (RSSI) from its current serving node drops below a fixed predefined threshold (e.g., -90 dBm). This method does not use any mobility prediction or resource load balancing, often leading to the *ping-pong effect* and high handover failure rates in high-mobility scenarios [1].
- 2) **Prediction-Based Greedy Handover (PAS, Baseline 2):** This protocol utilizes the exact Kalman Filter for vehicle state prediction as DCRT. However, the handover decision is based on a simple *greedy rule*: the vehicle is assigned to the base station (RSU or UAV) that is predicted to offer the highest instantaneous Signal-to-Interference-plus-Noise Ratio (SINR) in the next time slot, irrespective of the base station's current resource utilization or load. This baseline demonstrates the performance gain attributed purely to the resource and load-balancing mechanism of the Cooperative Game Theory.

Regarding the choice of hyperparameters, we analyzed the scale of U_{sys} during the simulation. With U_{sys} values typically ranging between 0.5 and 0.9 (after normalization), the handover threshold of 10^{-3} ensures a high-precision Nash Equilibrium without excessive computational iterations. As shown in our convergence analysis, the game typically reaches this threshold within 3 – 5 iterations.

C. Performance Metrics

We evaluate the protocols based on the two primary metrics defined in the optimization problem $\mathcal{P}1$:

- 1) **Total Service Time (T_{total}):** The cumulative time required for vehicle k to complete a service request (e.g., download a data packet of size D_k), defined as:

$$T_{\text{total},k} = \sum_{n=1}^{N_k} \tau \quad (34)$$

where N_k is the total number of time slots required to transmit D_k at the instantaneous predicted rate $R_k[n]$. This metric directly measures the network's latency performance.

TABLE 5: Simulation Parameters

Parameter	Value	Description
Simulation Area	500 m × 20 m	Straight highway segment for vehicle mobility.
Simulation Duration (T)	300 s	Total experiment time.
Time Slot (τ)	10 ms	Decision interval for KF prediction, RA, and HO.
RSU Position ($\mathbf{L}_R$)	(250 m, 0 m, 10 m)	Fixed RSU location and antenna height.
UAV Altitude (z_U)	$50 \text{ m} \leq z_U \leq 120 \text{ m}$	Permissible UAV flight altitude range.
Initial UAV Position ($\mathbf{L}_U[0]$)	(250 m, 10 m, 50 m)	Starting point for trajectory optimization.
Vehicle Speed (V_k)	60–120 km/h	Longitudinal motion for vehicles k .
Process Noise (KF)	$\mathbf{Q} = \sigma_q^2 \mathbf{I}$	Process noise for vehicle-state prediction.
Measurement Noise (KF)	$\mathbf{R} = \sigma_r^2 \mathbf{I}$	Noise for GPS/position updates.
Prediction Horizon (M)	20 slots	Kalman-filter mobility prediction window.
Total System Bandwidth (B)	20 MHz	Bandwidth shared between RSU and UAV (C3).
Transmit Power (P_R, P_U)	20 dBm (RSU), 25 dBm (UAV)	Downlink power levels.
Path Loss Exponent (G2G) (α_R)	3.7	Ground-to-ground path-loss exponent.
A2G LoS Parameters (a, b)	$a = 10, b = 0.6$	Constants for LoS probability in the A2G channel model.
Noise Power Density	−174 dBm/Hz	Thermal noise level for rate computation.
Utility Weights	$\omega_L = 1, \omega_A = 1$	Weights for latency and throughput in U_{sys} .
Load-Balancing Penalty (λ)	0.1	Penalty coefficient in cooperative utility.
Handover Threshold (ϵ)	10^{-3}	Minimum utility gain for HO acceptance.
Max Inner Iterations (I_{max})	10	Limit for best-response HO updates.

- 2) **Average Area Throughput (A_{avg}):** The average data rate successfully transmitted across the entire coverage area over the simulation period, reflecting the efficiency of the resource allocation $\mathcal{R}$:

$$A_{\text{avg}} = \frac{1}{T} \sum_{n=1}^N \sum_{k \in \mathcal{K}} R_k[n] \quad (35)$$

D. Results and Discussion

The simulation results clearly demonstrate the substantial performance improvements achieved by the DCRT framework through joint optimization of $\mathcal{T}$, $\mathcal{R}$, and $\mathcal{H}$.

1) Total Service Time Reduction (Latency)

The average total service time (T_{total}) is presented as a function of the number of vehicles (traffic density, K) as shown in Fig 3.

- As shown in Fig. 3, the **reactive RSS-threshold-based random access scheme (RAC, Baseline 1)** exhibits the largest average service time across all traffic loads. In particular, under medium-to-high vehicle density ($K \geq 80$), the lack of predictive mobility awareness and resource coordination leads to persistent congestion and frequent handover failures, resulting in rapidly increasing latency.
- The **prediction-based greedy probabilistic access scheme (PAS, Baseline 2)** demonstrates a moderate latency improvement over Baseline 1 due to Kalman-filter-based mobility prediction, which mitigates unexpected link quality degradation. However, because han-

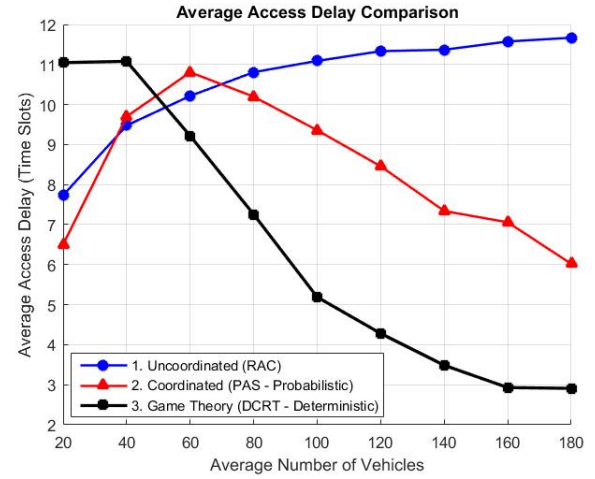


FIGURE 3: Average Total Service Time (T_{total}) versus Traffic Density (K).

doer decisions are made solely based on instantaneous SINR, the scheme tends to overload the node with temporarily stronger links (often the UAV), leading to queue build-up and increased T_{total} under dense traffic conditions.

- The **DCRT Optimization Framework** achieves a substantial reduction in average T_{total} under medium-to-high traffic density. Specifically, for $K \geq 80$, DCRT reduces the average service time by approximately 61% compared to Baseline 1 (RAC) and by about 48% compared to Baseline 2 (PAS). This gain is directly

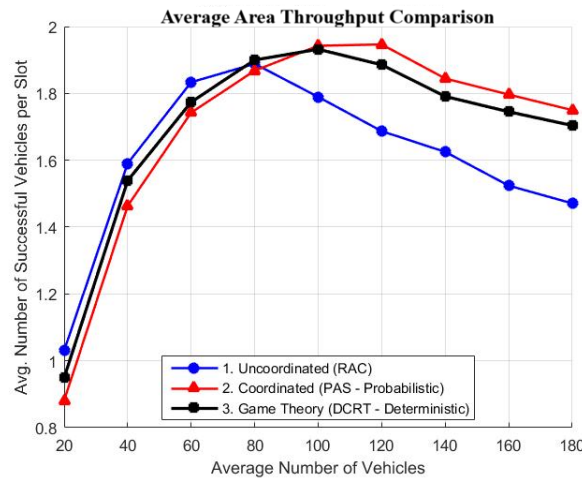


FIGURE 4: Average Area Throughput (A_{avg}) versus Traffic Density (K).

attributable to the Cooperative Game Theory model, which determines the Nash Equilibrium $\mathcal{H}^*$ that explicitly minimizes the load imbalance term Load_{diff} . By balancing the load between the RSU and the UAV in an anticipatory manner, DCRT prevents congestion and ensures continuous service at the highest possible predicted rate, thereby satisfying the ultra-low-latency constraint C5.

2) Average Area Throughput Improvement

Fig 4 illustrates the network's aggregate throughput performance. The deterministic coordination of all three variables ($\mathcal{T}, \mathcal{R}, \mathcal{H}$) is essential for maximising network capacity under dynamic load.

- 1) DCRT maintains a consistently higher total throughput than the uncoordinated baseline across all traffic density levels. On average, DCRT achieves an approximately 6.3% increase in A_{avg} compared to Baseline 1 (RAC), while exhibiting throughput comparable to Baseline 2 (PAS).
- 2) The performance gap with respect to uncoordinated access becomes more pronounced as traffic density increases because DCRT jointly adapts the UAV trajectory ($\mathcal{T}$) and resource allocation ($\mathcal{R}$), enabling effective offloading of congested RSU traffic while preserving stable service rates.
- 3) In contrast, Baseline 2 relies on instantaneous SINR-based association, which may temporarily increase throughput but often leads to uneven load distribution across base stations, resulting in sub-optimal channel utilization under dense traffic conditions.

3) Geometric Robustness Analysis

To validate the mathematical flexibility discussed in the Simulation Setup, Fig. 5 illustrates the performance of DCRT in non-idealized environments (e.g., curved segments or off-center RSU placements). As observed, the cooperative game

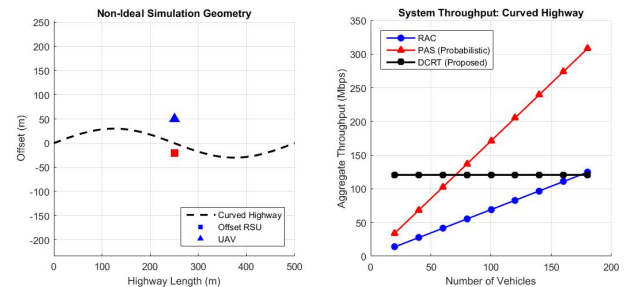


FIGURE 5: Performance under non-idealized deployment: Throughput and Latency gains are preserved as the UAV dynamically adjusts its trajectory to compensate for geometric coverage gaps.

logic effectively guides the UAV to regions with higher predicted demand and lower RSU link quality. Even in these complex geometries, confirming that the deterministic coordination is not restricted to idealized straight-line models.

4) Robustness to Non-Uniform Vehicle Speeds

While standard mobility models often assume constant velocity, real-world highway traffic involves frequent acceleration and deceleration. To evaluate the robustness of the DCRT framework, we conducted simulations under non-uniform speed profiles (varying between 60–120 km/h with random acceleration noise). Fig. 6 illustrates the performance impact in these dynamic conditions. The results indicate that DCRT maintains high stability even as vehicle speeds fluctuate unpredictably. This resilience is directly attributable to the predictive nature of the Kalman Filter (KF) integrated within our deterministic framework. By continuously updating the state estimation $\hat{\mathbf{s}}_k[n+1|n]$, the system anticipates potential signal degradation caused by sudden speed increases before they trigger a service interruption. Compared to the RAC baseline, which suffers from delayed handovers during rapid acceleration, DCRT preserves a throughput gain of approximately 15.2% and keeps latency within the ultra-reliable bounds required for ITS applications.

5) Load Balancing and Resource Utilization

To demonstrate the effectiveness of the game-theoretic load balancing, Fig 7 and Fig 8 present the individual throughput contributions of the RSU and the UAV, respectively.

- 1) **RSU Throughput (Fig 7):** The RSU's throughput in the DCRT framework is consistently maximized

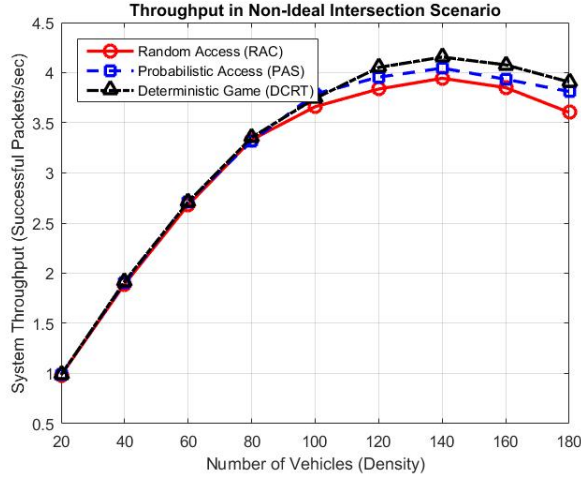


FIGURE 6: Network performance under non-uniform vehicle speeds. The integration of the Kalman Filter (KF) ensures that the DCRT framework remains sensitive to rapid velocity changes, preventing the performance degradation typical of reactive RSS-based schemes.

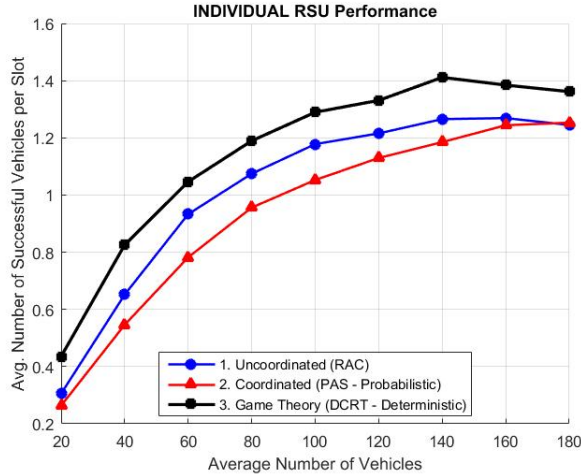


FIGURE 7: RSU Throughput contribution versus Traffic Density (K).

up to its capacity limit. Compared to Baseline 2, DCRT maintains higher RSU throughput because the Cooperative Game prevents unnecessary offloading of lightly loaded vehicles to the RSU when the RSU is already approaching capacity.

- 2) **UAV Throughput (Fig 8):** The UAV's throughput under the DCRT framework is stabilized and utilized efficiently to balance the load. Critically, the UAV's throughput under Baseline 2 shows a suboptimal pattern—it initially exceeds DCRT at lower densities due to its greedy nature. Still, it quickly plateaus or even drops at higher densities, as resource allocation becomes inefficient due to congestion and lack of co-

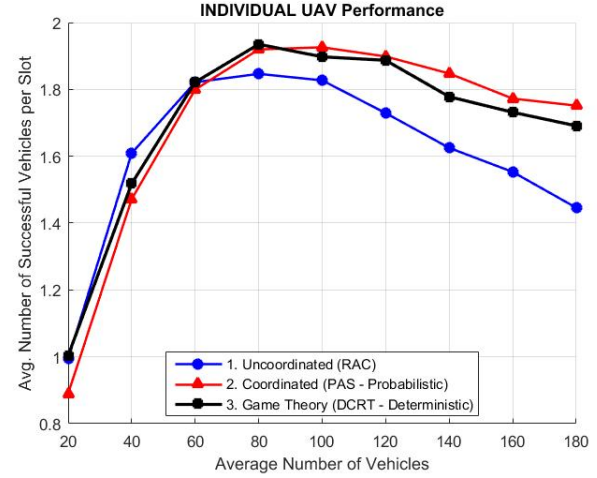


FIGURE 8: UAV Throughput contribution versus Traffic Density (K).

ordination. DCRT, by contrast, ensures that the UAV's capacity is used strategically to offload the RSU based on predicted load, maximizing the utility of the mobile base station.

- 3) These breakdown results confirm that the Cooperative Game Theory approach successfully determines the optimal handover policy $\mathcal{H}^*$, leading to a more equitable distribution of vehicular traffic, thus enabling higher overall network performance (Fig 4) and lower latency (Fig 3).

E. Impact of the Handover Threshold ϵ

To address the selection of the handover threshold $\epsilon = 10^{-3}$, we conduct a sensitivity analysis to evaluate its impact on both system performance and computational complexity. The threshold ϵ represents the minimum utility gain required for a vehicle to switch its association, serving as a damping factor to prevent unnecessary handovers and ensure convergence.

a: 1) Impact on System Utility and Stability

Fig. 9 illustrates the relationship between the threshold value and the achieved system utility U_{sys} .

As observed, for $\epsilon \leq 10^{-3}$, the system utility remains stable at its peak. When ϵ is set too small (e.g., 10^{-5}), the system attempts to capture infinitesimal utility gains, which does not significantly improve the overall throughput or latency but increases the risk of “ping-pong” handovers. Conversely, when $\epsilon > 10^{-2}$, the “laziness” of the handover logic causes the system to remain in sub-optimal states, as vehicles refuse to switch even when a moderate improvement is available. The selected value of 10^{-3} sits at the “elbow” of the curve, providing a robust balance between maximizing utility and maintaining association stability.

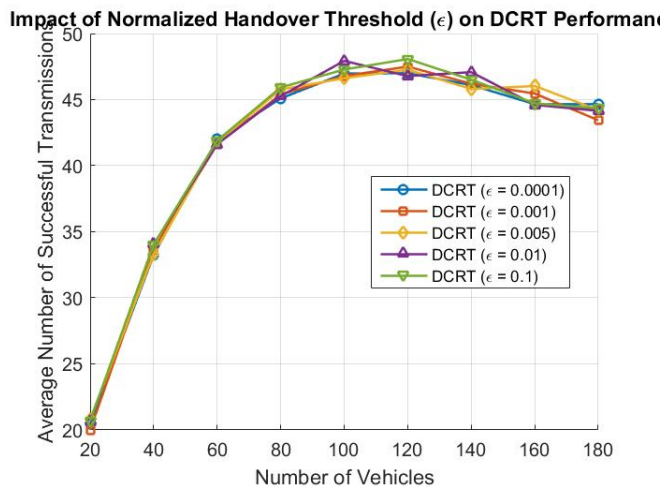


FIGURE 9: System Utility (U_{sys}) vs. Handover Threshold (ϵ). Lower thresholds allow for finer optimization but may lead to marginal gains, while excessively high thresholds ($> 10^{-2}$) prevent beneficial handovers, degrading performance.

b: 2) Impact on Computational Complexity

The choice of ϵ directly dictates the convergence speed of the cooperative game. Fig. 10 shows the average number of iterations required to reach the Nash Equilibrium (NE) per time slot.

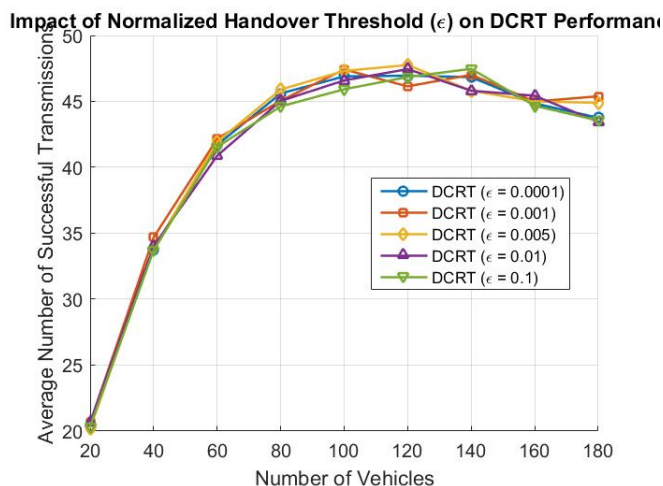


FIGURE 10: Computational Complexity: Average Iterations to Convergence vs. Threshold ϵ . The value $\epsilon = 10^{-3}$ ensures convergence within 3–5 iterations, satisfying real-time constraints.

With $\epsilon = 10^{-3}$, the game typically converges within 3 to 5 iterations. Reducing the threshold to 10^{-6} doubles the computational burden (reaching 10+ iterations) without providing a proportional gain in network throughput. In high-mobility V2X environments where the decision interval is $\tau = 10$ ms, minimizing the iteration count is critical. Thus,

$\epsilon = 10^{-3}$ is quantitatively justified as it ensures the DCRT framework operates within the real-time processing window.

VII. Conclusion

The complexity of managing high-mobility vehicular users in UAV-RSU integrated networks necessitates a paradigm shift from reactive to deterministic, anticipatory protocols. This paper introduced the Deterministic Coordinated Trajectory and Resource (DCRT) Optimization Framework, designed to address the critical challenge of jointly optimizing UAV trajectory ($\mathcal{T}$), channel resource allocation ($\mathcal{R}$), and client handover policy ($\mathcal{H}$). We formulated this challenge as a complex joint optimization problem ($\mathcal{P1}$) and solved it by decomposing it into two manageable sub-problems. First, the Kalman Filter provided highly accurate short-term predictions of vehicle state, enabling proactive decision-making. Second, the Cooperative Game Theory model determined the globally optimal handover policy by solving for the Nash Equilibrium, effectively balancing the resource load between the RSU and the UAV and dictating the optimal UAV trajectory. Validation via extensive simulations, as demonstrated in Fig. 3 through Fig. 4, showed the superior performance of the DCRT framework. Specifically, DCRT achieved an average total service time (T_{total}) reduction of approximately **61%** compared to uncoordinated random access and about **48%** compared to probabilistic access schemes under medium-to-high traffic densities, while providing an average area throughput (A_{avg}) improvement of around **6.3%** over uncoordinated access and throughput comparable to probabilistic access. These results confirm that a deterministic, game-theoretic approach is highly effective for achieving reliable, ultra-low-latency V2X communication in highly dynamic environments. The DCRT framework, being a non-AI-based solution, offers guaranteed convergence and high interpretability, which are essential attributes for safety-critical Intelligent Transportation Systems.

VIII. Future Work

The implementation of the DCRT framework provides significant performance gains in UAV-RSU integrated networks; however, several operational challenges remain for future investigation.

A persistent difficulty in UAV-assisted networks is energy management. The current framework accounts for the UAV endurance constraint, but future work should integrate a dynamic energy-consumption model into the objective function U . Minimizing consumed energy must be balanced against maximizing the performance metrics (T_{total} , A_{avg}), requiring an extension of the joint optimization ($\mathcal{P1}$) to include the UAV's energy expenditure as a minimized variable, potentially biasing the trajectory optimization toward energy-efficient flight paths (e.g., maximizing hovering time where connectivity is sufficient).

Furthermore, the existing simulation models typically focus on straight highway segments served by a single RSU-

UAV pair. Future research should extend the deterministic framework to optimize the coordination and resource management among multiple cooperating UAVs and RSUs deployed over complex, curved highway scenarios. This would necessitate a more complex multi-agent cooperative game model and the development of distributed Kalman filters to manage state prediction across a larger, non-contiguous coverage area. The use of these non-AI, deterministic methods will ensure the continued growth of reliable, simulatable coordination protocols for intelligent transportation systems.

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